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Ideal Homes Township, RajaRajeshwari Nagar, Bengaluru-98

SMART IOT MINIPROJECT

PROJECT REPORT

Title: Ultrasonic Distance Measurement System

Submitted by: Tejas S
USN: 1GA21CS167

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1. Introduction

Accurate distance measurement plays a vital role in fields such as robotics, automation, and security systems. Ultrasonic sensors provide a dependable and cost-effective solution for high-precision distance measurements. This project involves designing and implementing an ultrasonic distance measurement system using an Arduino (or ESP32) microcontroller along with an HC-SR04 ultrasonic sensor. The system calculates and displays the object's distance from the sensor in real time.

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2. Objectives

- Develop an ultrasonic distance measurement system utilizing an Arduino/ESP32.

- Integrate an ultrasonic sensor for real-time distance detection.
- Display the measured distance via a serial monitor.
- Ensure precise and efficient readings for diverse applications.

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3. Components Required

- Microcontroller – Arduino/ESP32
- Ultrasonic Sensor – HC-SR04
- Resistors and Connecting Wires – For circuit assembly
- Power Source – 5V DC supply
- Software – Arduino IDE

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4. Circuit Diagram

The circuit connects the HC-SR04 ultrasonic sensor to the microcontroller. The Trigger (TRIG) pin is linked to GPIO 5, while the Echo (ECHO) pin connects to GPIO 18. The sensor emits ultrasonic pulses, which reflect off an obstacle and return to the sensor, allowing distance calculations.

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5. Working Principle

The system operates based on the time-of-flight concept. It transmits an ultrasonic pulse, which reflects from an object and returns to the sensor.

Distance=Time×0.0342\text{Distance} = \frac{\text{Time} \times 0.034}{2}Distance=2Time×0.034

where 0.034 represents the speed of sound in cm/µs, and the division by 2 accounts for the round-trip travel time.

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6. Code Implementation

```
// Pin definitions

#define TRIG_PIN 5  // Define the GPIO pin for the trigger

#define ECHO_PIN 18  // Define the GPIO pin for the echo


long duration; int

distance;


void setup() {

Serial.begin(115200);

pinMode(TRIG_PIN, OUTPUT);

pinMode(ECHO_PIN, INPUT);

  Serial.println("Ultrasonic Sensor Initialized.");

}


void loop() {   digitalWrite(TRIG_PIN,

LOW);   delayMicroseconds(2);

digitalWrite(TRIG_PIN, HIGH);
```

```
delayMicroseconds(10);
digitalWrite(TRIG_PIN, LOW);

  duration = pulseIn(ECHO_PIN, HIGH);
distance = duration * 0.034 / 2;

  Serial.print("Distance: ");
  Serial.print(distance);
Serial.println(" cm");   delay(1000);
}
```

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7. Results and Observations

- The system effectively measures distances in real time with accuracy.
- The measured distance is displayed in centimeters on the serial monitor.
- The sensor provides quick response times, ensuring efficiency.
- While the system works well indoors, outdoor use may require calibration.

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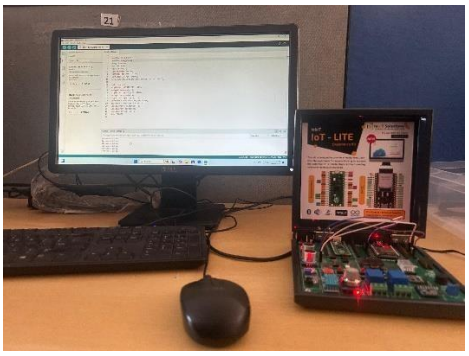
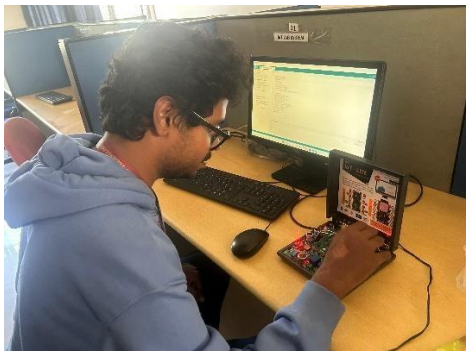
8. Applications

- Robotics: Assists in obstacle detection and navigation.
- Security Systems: Helps in intruder detection for restricted areas.
- Smart Vehicles: Supports parking assistance and collision avoidance.
- Industrial Automation: Measures object distances in manufacturing setups
- 



9. Conclusion

This project successfully demonstrates an ultrasonic distance measurement system using an HC-SR04 sensor and a microcontroller. The system is reliable, efficient, and applicable across various fields such as automation, robotics, and security. Future improvements could include integrating an LCD display, wireless data transmission, or multiple sensors to enhance accuracy.

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10. References

- Datasheet of HC-SR04 Ultrasonic Sensor
- Arduino Documentation
- Online tutorials on ultrasonic distance measurement

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